

AeroNetB Aerospace Supply Chain Management

Database Design Document

Field	Value
Module	5CM506 Data Driven Systems
Student Name	Spyros Kokkoris
Student ID	100774175
Submission Date	May 2026

1. Introduction

AeroNetB Aerospace manufactures critical commercial aircraft components including fuselage sections and wing assemblies. The company relies on a global supplier network where each supplier may provide the same baseline part with different supplier-specific customisations, inspection evidence, shipment events, and certification records.

The system replaces fragmented spreadsheets, legacy databases, and manual logs with one data-driven platform for supplier management, order tracking, shipment monitoring, QC reporting, certification control, IoT telemetry, role-based access, and audit logging.

1.1 System Objectives

Objective	Description
O1	Maintain central supplier, part, and order records with relational integrity.
O2	Model supplier-specific part variability without duplicating baseline part data.
O3	Store flexible QC reports, certification payloads, and IoT readings in document collections.
O4	Enforce RBAC and audit logging across the API and dashboard.
O5	Provide a live Render-hosted dashboard backed by PostgreSQL and MongoDB.

2. Requirements Analysis

Requirement	Description
Supplier and part management	Suppliers have accreditation and contact information. Parts have baseline specifications. Supplier-part offerings store supplier part codes and customisation notes.
Orders and shipments	Purchase orders track status from placed to completed. Shipments store tracking numbers, ETA, ports of entry, and event updates.
Quality control	QC reports vary by inspection type. Reports are versioned and linked to supplier, part, order, and inspector identifiers.
Certification	Certification documents contain test results, batch origin, digital signature, and approval metadata. Approved records become immutable.
IoT monitoring	Equipment and transit devices produce timestamped sensor readings including temperature, vibration, pressure, and GPS data.
RBAC and audit	Users authenticate by auth ID. Permissions control read, write, approve, export, and audit actions. Important actions are logged with emp_id.

3. Conceptual Design

The conceptual model is technology-agnostic. It identifies the main business entities, attributes, and relationships. Dashed entities in the figure represent document-style domains linked through shared identifiers.

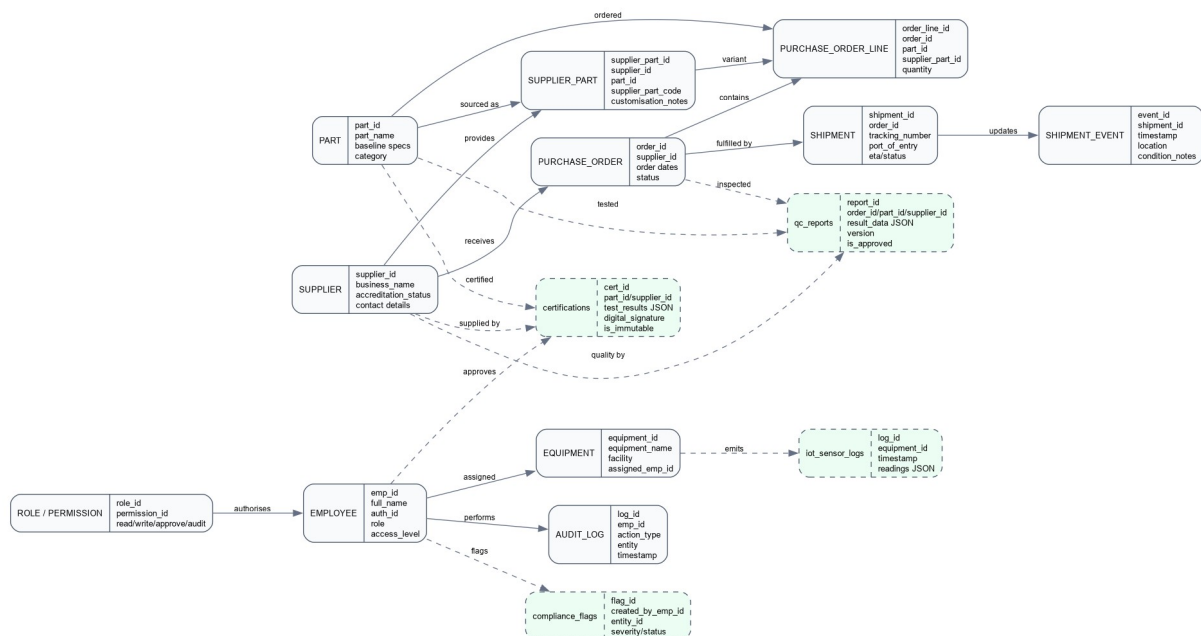


Figure 1 - Conceptual ER Model for AeroNetB ASCM

3.1 Key Modelling Choices

Choice	Justification
Separate Part from Supplier_Part	The same baseline aerospace part can be sourced from multiple suppliers, but each supplier can add different features.
Use Shipment_Event	Shipment location and condition updates are repeated over time, so they belong in a separate event entity.
Use Employee, Role, Permission, and Audit_Log	The scenario requires every important action to be controlled by role and attributable to emp_id.
Use MongoDB for QC, certification, and IoT payloads	These domains contain flexible nested fields, varying measurement structures, and high-volume telemetry.

4. Logical Design

4.1 Relational Logical Model

The relational model is implemented in PostgreSQL. It stores structured master data, transaction data, identity data, role data, equipment records, and audit events.

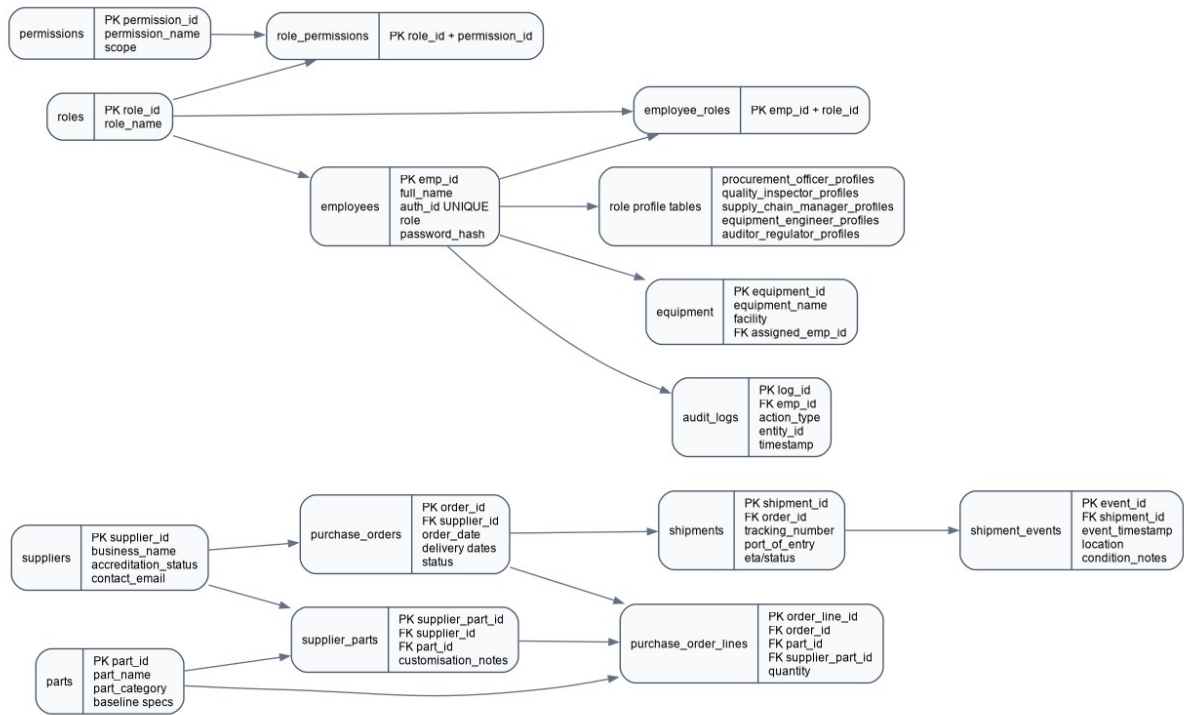


Figure 2 - PostgreSQL Relational Logical Model

4.2 Document Logical Model

MongoDB stores flexible document data that changes shape or grows continuously. Documents retain shared IDs from PostgreSQL so the API can join data for dashboard responses.

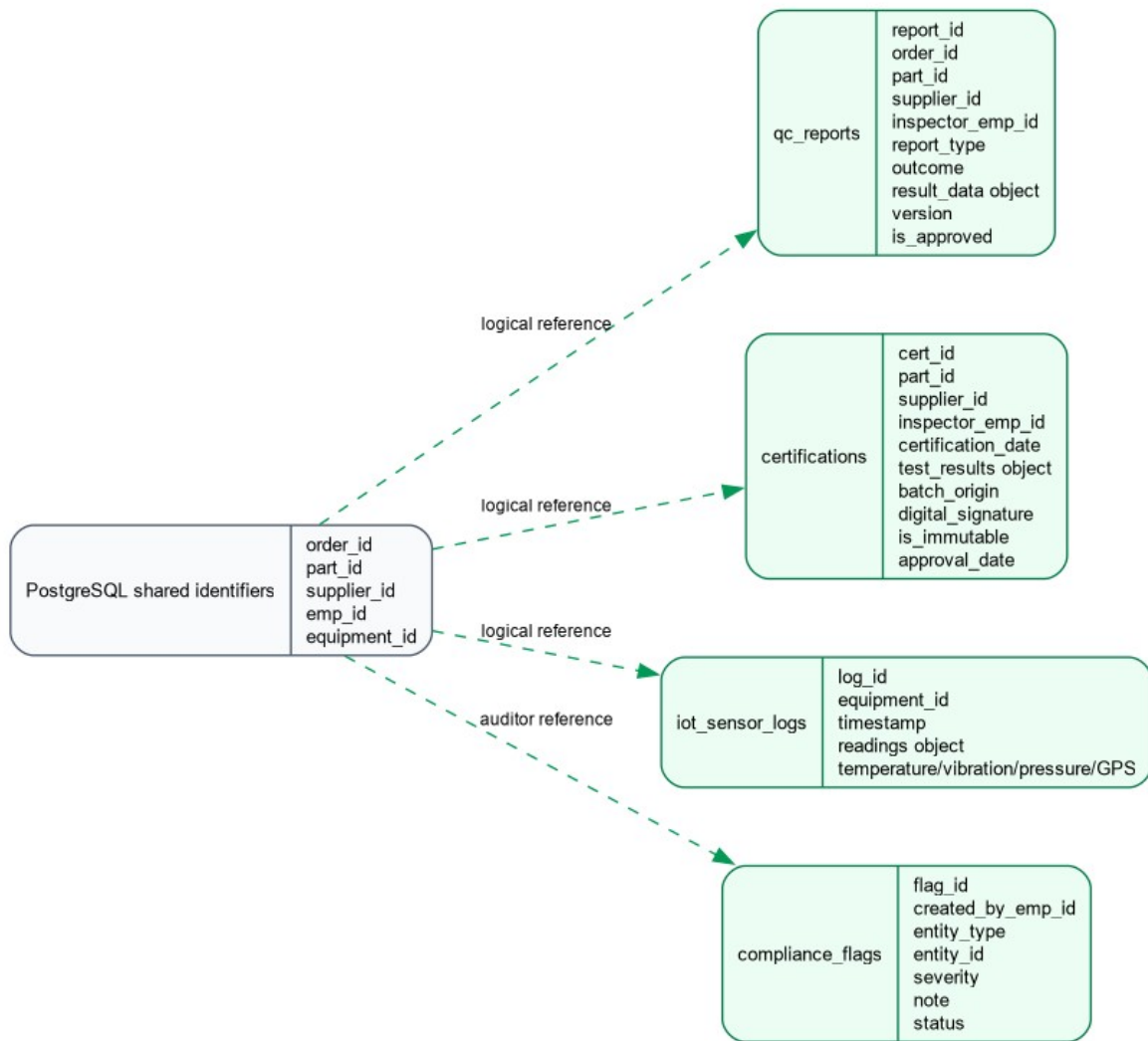


Figure 3 - MongoDB Document Logical Model

4.3 Cross-Database Consistency

PostgreSQL is the authoritative source for stable identifiers and relational integrity. MongoDB documents store those identifiers as logical references. The API checks relational records before creating QC reports, certifications, or IoT logs. Certification immutability and QC versioning are enforced in the backend service.

5. Conclusion

The final design satisfies the AeroNetB scenario by combining a normalised PostgreSQL schema with flexible MongoDB document collections. The model supports supplier-part variability, order and shipment lifecycle tracking, quality reporting, certification traceability, IoT monitoring, RBAC, and audit logging.

References

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